Syslog

GRAFAN LOKI, ALLOY, AND NGINX

What is Log Management and Syslog

Log management: Process of centrally collecting, parsing, and store log data

- Useful for tracking down errors across host
- Required for certain compliance standards

Syslog: standard messaging protocol for applications to generate messages for log management

- Runs on port 514 commonly on UDP but can also be TCP depending on how critical the logs are

Message contains the following parts:

- Facility: what type of system generate the message
- Severity: from 0 emergency to 7 debug
- Message: contains a tag with name of program/process and content with the acutal log message

Why Grafana Loki and Alloy

Multiple different syslog collection and management. More common to use other solutions like Splunk, Graylog, or ELK stack

Grafana Loki: lightweight in terms of storage and process comes at the cost of searching

- Only the tag is indexing not the entire log so searching based on a word will take longer
- Searching is also more limited in what you can do

Grafana Alloy: Sits between the application and log server, collects and forwards logs

Grafana Repo and Other Packages

Go to Grafana website for the commands to add the Grafana repo and install Grafana OSS at https://grafana.com/docs/grafana/latest/setup-grafana/installation/debian/

Then install the other packages necessary: sudo apt install vim loki alloy nginx wireshark

Setting Up Grafana Loki

1. Start the service: sudo systemctl start loki
2. Check the service status: systemctl status loki
3. If it reports an error check the journal with sudo journalctl -xeu loki
4. If the error is failed to determine user credentials, then create the loki system account with sudo useradd -M -r -s /sbin/nologin loki
5. If you had to create the account, then restart loki: sudo systemctl restart loki

Setting Up Grafana

1. Check the Grafana service: systemctl status grafana-server
2. If it says inactive then start the service: sudo systemctl start grafana-server
3. Go to Grafana through the browser at localhost:3000 and login with admin admin
4. Change the password to something that you will remember
5. Add the Loki source by going to connections on the left menu then add new connection
6. Search for Loki then click on it then click on add new data source in top right
7. Set url to http://localhost:3100, then scroll down to the bottom and click save and test

Configuring Alloy

1. Open terminal and edit the config.alloy with sudo vim /etc/alloy/config.alloy
2. Add the following at the bottom of the file

```
loki.source.syslog "nginx" {
   listener {
           address = "127.0.0.1:1514"
           protocol = "udp"
           syslog_format = "rfc3164"
           labels = {
                   job = "nginx",
           }
   }
   forward_to = [loki.write.local.receiver]
}
loki.write "local" {
   endpoint {
       url = "http://127.0.0.1:3100/loki/api/v1/push"
   }
}
```

Configuring Alloy

- Save and exit the file then check the configuration: sudo alloy validate /etc/alloy/config.alloy
- If no errors restart alloy: sudo systemctl restart alloy

Configuring Nginx

1. Edit the Nginx default site: sudo vim /etc/nginx/sites-enabled/default
2. Add the following in the http block:
   error_log syslog:server=127.0.0.1:1514;
   access_log syslog:server=127.0.0.1:1514;
3. Save and exit then check the configuration: sudo nginx –t
4. If it was successful reload nginx: sudo systemctl reload nginx
5. Go to the browser and go to localhost to generate an acess log

View the Logs

1. Go back to Grafana at localhost:3000 -> explore -> then Loki
2. Switch from builder to code in the query and put in {job="nginx"}
3. Click run query, the blue two arrow circle button in top right
4. You should see something like the below



Viewing with Wireshark

Since we are using HTTP on Grafana Alloy and Loki we can view the packet in Wireshark

1. Open Wireshark from the terminal with sudo wireshark and click on loopback
2. In the browser access localhost a few times
3. Look for some HTTP traffic followed soon after by UDP traffic
4. Click on the UDP traffic, on the right side of the packet inspection you can see the decoded log message that has the same information as the log in Grafana Loki